

# OM CHIMURKAR

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## Education

**Newton School of Technology, Rishihood University**  
*B.Tech in Artificial Intelligence*

*Expected May 2028*  
**GPA: 8.0 / 10.0**

## Research & Publications

### DKV: Anchor + Low-Rank Differential KV-Cache Compression

*IEEE TETCI (Under Review) | Technical Report & Open-Source Runtime | Zenodo: 10.5281/zenodo.21539110*

*Independent Research*

- Architected a training-free KV-cache compression runtime combining exact anchors, rank-32 joint KIV truncated SVD deltas, and multi-signal exact residual tokens to preserve verbatim recall.
- Achieved 1.44x–2.25x memory reduction and 1.72x faster prefill scaling at 64k context on Apple M3 (8.6 GB) over memory-optimized dense baselines.
- Engineered MLX and CUDA/Triton backends with autotuned fused decode launches, async H2D prefetching, and in-place residual scattering; exact residual keys prevented needle-recall collapse across 4k–64k contexts with  $\Delta\text{PPL} = 0.00\%$ .

### Deterministic AGI Benchmarking Under Constructed Cognitive Load

*Published Preprint | Zenodo: 10.5281/zenodo.19652443*

*Independent Research*

- Designed a deterministic benchmark using constructed-language rule-learning tasks to evaluate reasoning without judge-model scoring, reducing judge inflation and memorization effects.
- Developed 116 original tasks across Legal, Science, Social Norms, and Math/Logic; evaluation across seven frontier models revealed a 23.2 percentage-point performance spread.

### An Ablation Study of Retrieval Strategies for Stock-News RAG Systems

*Published Preprint | Zenodo: 10.5281/zenodo.19086005*

*Independent Research*

- Conducted controlled empirical ablations comparing semantic, hybrid (BM25 + vector), and Maximal Marginal Relevance (MMR) retrieval on stock-news QA benchmarks.
- Demonstrated that hybrid retrieval improves coverage over semantic-only search, while MMR increases contextual diversity without proportional grounding gains.

## Technical Skills

**LLM & ML Systems** PyTorch, Hugging Face Transformers, LLM Inference Optimization, KV-Cache Compression, RAG, Model Evaluation, Quantization, Low-Rank Factorization (SVD)

**GPU & Software Systems** Python, CUDA, Triton, Nsight Systems, Nsight Compute, FastAPI, Git, Vector Databases, Linux

## Awards & Experience

### Cognizance '25 (SnapSyntax) — IIT Roorkee

**1st Place / Zonals Winner**

*May 2025*

- Secured first place in the SnapSyntax technical challenge at IIT Roorkee's annual technical festival.

### Google — Gemini API Developer Contest

**Contestant**

*May–August 2024*

- Built and deployed an asynchronous LLM application integrating Gemini API with custom prompt pipelines under live competition constraints.

## Honors & Certifications

**Anthropic:** AI Fluency Framework & Foundations

**AWS:** Fundamentals of Machine Learning & Artificial Intelligence